# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**Bachelor of Technology (B.Tech)**

**Semester – I / II University Examination May 2012**

**B.Tech. (CE/EE/IT) / B.Tech. (CL/EC/ME)**

## CE – 101 Fundamentals of Computing & Programming

**Date: 16/05/2012, *Wednesday*** **Time: 10:00 a.m to 1:00 p.m** **Maximum Marks:70**

**Instructions:**

*1. Attempt all questions.*
*2. Answers both sections in separate answer books.*
*3. Figure to the right indicates full marks.*
*4. Assume suitable data, if necessary and mention it.*

## SECTION – I

**Q.1(a) Answer the following questions.** **05**

1. Which is the longest data-type in C programming language on a 16-bit machine?
2. Differentiate: *Entry Controlled loop and Exit Controlled loop.*
3. What are the differences between compiler and interpreter?
4. Explain the purpose of ternary operator.
5. What is the use of sizeof operator?

**Q.1(b) Do as directed.** **06**

(1)
What will be the output of the above C program?

```
void main ( )
 {
 int A=15,B=15,C;
 C=A++ + ++B + --B;
 printf("%d",C);
 }
```

(2)
What is the size of the following structure?

```
struct class
{
  char name[20];
  int number;
  float marks;
}
```

(3)
Is there any error present in the following program? If yes, correct it. Write down the output.

```
void main()
{
  switch(1+1)
{
case 1:
        printf("Hello");
case 2:
        printf("Good");
case 3:
        printf("Morning");
}}
```

**Q.2 Answer the following questions.** **12**

1. Explain: Structure of C programs.
2. What is the difference between array and structure? Explain with an example.
3. Write short note on: *Type Conversions in Expressions.*

**OR**

**Q.2 Answer the following questions.** **12**

1. List out arithmetic operators in C. Explain any three of them in detail.
2. Which are the various operations that can be performed on a file? Also explain *fopen(), fclose() and fseek()* functions.
3. What do you mean by Recursion? When is it used? Give an example for it.

**Q.3 Answer the following questions.**

1. Write a program to calculate the sum and average of a set of N numbers. **06**
2. Write a function using pointers to exchange the values stored in two locations in the memory. **06**

**OR**

**Q.3 Answer the following questions.**

1. Write a program to make simple calculator using arithmetic operator as input using switch…case. **06**
2. Differentiate: *printf()* and *scanf()* functions. Also write down a C program to that converts the given temperature in Fahrenheit to Celsius using the following conversion formula: C=(F-32)/1.8 **06**

## SECTION – II

**Q.4 Do as directed.** **11**

1. Is main () a user defined function? If yes how does it differ from other user-defined function? If No, give reason for it. **03**
2. Find out the value: $(AB)_{16} = (?)_{10}$ and $(1111001)_2 = (?)_8$ **02**
3. Evaluate the following expression with steps according to C Programming. **02**
   2*3/4+4/4+8-2+5/8
4. Classify the following variables into valid and invalid variable name in C. **02**

| | | |
|---|---|---|
| % | (area) | mark |
| 25th | 123 | distance |

5. Match the following: **02**

| A | B |
|---|---|
| 1. #define | 1. Global variable known to all function in the file |
| 2. extern | 2. Preprocessor Compiler Directive |
| 3.getchar() | 3. Concatenates two strings |
| 4.strcat() | 4. Accepts any character keyed in |

**Q.5 Answer the following questions.** **12**

1. Explain one-dimensional arrays in details.
2. Explain strcmp() and strcpy() with example.
3. Comment on this statement:" Use of goto should be avoided". What do you mean by forward jump and backward jump used in goto statement?

**OR**

**Q.5 Answer the following questions.** **12**

1. Explain various categories of functions.
2. Differentiate: Call by value and Call by reference with an example.
3. Explain: Copying and Comparing Structure Variables using an example.

**Q.6 Write down following programs in C programming language.**

1. Write a C program to find factorial of a given number using function concept. Make fact () as a user defined function. **06**
2. Write a C program to compute the real roots of a quadratic equation: $ax^2+bx+c=0$. **06**

**OR**

**Q.6 Write down following programs in C programming language.**

1. Define a structure that describes a student having members roll no, name, mark1, mark2, mark3. Write a program for five students to print the total of marks and average of marks. **06**
2. Write a C program to generate Fibonacci series. **06**

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